Project Status Report

Project Title: Privacy-Preserving Federated Learning Framework for Multimodal Pneumonia Diagnosis

Work Completed So Far

Over the past week, I focused on refining the direction and scope of our final-year project through literature survey, technical discussions, and domain analysis. Initially, our team identified the broader problem of applying **Federated Learning in Healthcare**, with the objective of enabling multiple hospitals to collaboratively train artificial intelligence models while preserving patient privacy. To better understand real-world healthcare requirements, we conducted a survey and discussions at **B & B Hospital**. This helped us understand the practical challenges of medical data sharing and reinforced the need for a privacy-preserving collaborative learning framework. Based on these findings, we designed the overall system architecture of our proposed framework. The architecture includes local hospital training, a central aggregation server, global model aggregation, and privacy-preserving mechanisms such as **Differential Privacy** and **Secure Aggregation**. We also prepared and presented our initial project abstract presentation based on this architecture. After reviewing recent research papers, we concluded that building a framework for general pulmonary disease diagnosis would significantly increase the project complexity. Therefore, we narrowed the application domain to **multimodal pneumonia diagnosis**, allowing us to develop a more focused and practical solution while retaining the Federated Learning framework. In parallel, we started discussions for the **Intermediate Project Presentation** and refined the following sections: Introduction Problem Definition Objectives Functional and Non-Functional Requirements Overall System Architecture Modules and Their Details Design Diagrams (Use Case, Activity Diagram, Sequence Diagram, Data Flow Diagram, etc.) This provided a clear roadmap for both the documentation and implementation phases of the project.

Plan for the Next Week

During the coming week, my primary focus will be preparing and presenting my seminar titled **“A Study on Transformer-Based Multimodal Learning for Pulmonary Disease Diagnosis.”** I plan to study the selected base paper and supporting research papers in detail to understand the transformer-based multimodal architecture, methodology, attention-based feature fusion, and experimental results thoroughly. Alongside the seminar preparation, our team plans to finalize and present the **Intermediate Project Presentation** by completing the system architecture, module descriptions, and design diagrams. We also intend to prepare a detailed implementation roadmap for the proposed Privacy-Preserving Federated Learning Framework for Multimodal Pneumonia Diagnosis.